On the Forecasting of Record-Breaking Extremes: Benchmarking Probabilistic AI Weather Forecasts

Author list to be finalized[1]

[1]Affiliations to be finalized

Abstract

AI-based weather forecasting now rival traditional numerical weather prediction in terms of average skill. Yet their reliability for record-breaking extremes remains unclear. Here, we extend the deterministic record-extreme benchmark of Zhang et al. [2026] to probabilistic ensemble forecasts of record-breaking heat, cold, and wind. In particular, we evaluate 2022 probabilistic forecasts of local record-breaking heat, cold and wind from IFS-ENS and three AI ensembles, FGN, U-Cast and ArchesWeatherGen, relative to monthly grid-cell-specific ERA5 records from 1979–2018. Using record-occurrence, distributional and margin-conditioned diagnostics, including Brier scores, reliability, precision-recall performance, record-conditioned continuous ranked probability scores (CRPS), ensemble coverage and out-of-ensemble rates, we find that record extremes are substantially harder to forecast than average weather. AI ensembles generally provide more skilful record-occurrence probabilities, particularly for heat, but this advantage does not consistently extend to the intensity of realised extremes: IFS-ENS remains competitive for cold and wind records, especially at longer lead times. As record margins increase, ensembles can retain probability of threshold exceedance while losing coverage of the realised intensity. Credible probabilistic forecasts of record extremes therefore require both calibrated occurrence probabilities and adequate tail support.

1 Introduction

Reliable forecasts of record-breaking extremes are essential for early warning, public health, energy systems, agriculture, transport and disaster response. Their importance has been underscored by the ongoing 2026 heat extremes: Copernicus reported that western Europe experienced its hottest June on record during an intense regional heatwave, while the month ranked as the second warmest globally. At the same time, WMO documented concurrent heat impacts, wildfire risk and record temperatures across multiple countries [Copernicus Climate Change Service (C3S), 2026, World Meteorological Organization, 2026]. A central forecasting question is therefore whether modern forecast systems remain credible when the atmosphere produces conditions beyond the local seasonal range represented in historical data.

Recent AI weather prediction systems have achieved strong global skill at substantially lower computational cost than traditional numerical weather prediction in several benchmark settings, including GraphCast, Pangu-Weather and FuXi [Rasp et al., 2024, Lam et al., 2023, Bi et al., 2023, Chen et al., 2023]. However, strong average skill does not guarantee skill for the tails of the distribution. A model can reduce global error by forecasting common conditions accurately while remaining unreliable for rare, high-impact extremes, where forecast performance is most consequential for early warning.

Very recently, Zhang et al. [2026] made this limitation explicit. They defined local record-breaking heat, cold and wind relative to each grid cell and calendar month, and assessed whether deterministic forecasts could exceed the corresponding historical record. ECMWF's physics-based deterministic forecast, HRES, outperformed GraphCast, Pangu-Weather and FuXi across all three event types. The AI forecasts underestimated hot and wind records and overestimated cold records, with biases increasing as the observed margin beyond the historical record grew. This behaviour is consistent with a "soft cap", whereby models trained on historical reanalysis interpolate accurately within the training distribution but struggle to extrapolate to unprecedented conditions.

The corresponding question for probabilistic forecasting remains open. Because weather is inherently chaotic and high-impact decisions require uncertainty quantification, forecast quality cannot be assessed from point forecasts alone. Probabilistic AI systems, including GenCast, FGN, U-Cast, ArchesWeatherGen, FuXi-ENS and AIFS-CRPS, produce ensembles or predictive distributions and are commonly trained using objectives related to proper scoring rules [Price et al., 2025, Alet et al., 2026, Cachay et al., 2026, Couairon et al., 2026, Zhong et al., 2025, Lang et al., 2026]. Such systems may assign probability beyond a historical record even when their ensemble mean is conservative. However, they may still be underdispersive, poorly calibrated in the tails, or fail to provide support at the realised intensity of unprecedented events.

Here, we extend the deterministic record-extreme benchmark to probabilistic ensemble forecasts. We evaluate the physics-based IFS-ENS alongside the AI ensembles FGN, U-Cast and ArchesWeatherGen using point-forecast summaries, record-occurrence diagnostics, continuous ranked probability score (CRPS), and margin-conditioned diagnostics of tail support and coverage.

2 Data and methods

Monthly local records are derived from ERA5 for the 1979–2018 reference period. We consider 2-m temperature and 10-m wind speed: heat records are upper records of 2-m temperature, cold records are lower records of 2-m temperature, and wind records are upper records of 10-m wind speed. The analysis is conducted on a 1.5° grid spanning approximately 60°S–90°N, excluding Antarctica, and is restricted to valid times at 00 and 12 UTC. Forecasts are evaluated for 2022.

Let $x_{s,t}$ denote a weather variable at grid cell s and verification time t. For calendar month m, the local historical upper and lower records over the reference period T_{ref} = 1979–2018 are

$$r^{x,+}_{s,m} = \max_{t \in T_{\text{train}}: m(t)=m} x_{s,t}, \tag{1}$$

and

$$r^{x,-}_{s,m} = \min_{t \in T_{\text{train}}: m(t)=m} x_{s,t}. \tag{2}$$

An upper-tail record in the test year occurs at (s,t) if

$$y^{+}_{s,t} = \mathbb{1}\{x_{s,t} > r^{x,+}_{s,m(t)}\} = 1, \tag{3}$$

whereas a lower-tail record occurs if

$$y^{-}_{s,t} = \mathbb{1}\{x_{s,t} < r^{x,-}_{s,m(t)}\} = 1. \tag{4}$$

Thus, record events are grid-time pairs whose verifying values exceed the local, month-specific historical range.

We evaluate ECMWF's physics-based IFS-ENS and three probabilistic AI systems: FGN, ArchesWeatherGen and U-Cast [Alet et al., 2026, Couairon et al., 2026, Cachay et al., 2026]. FGN

combines a graph-based architecture with learned functional perturbations and a deep ensemble, and is trained to optimize CRPS. ArchesWeatherGen generates residual uncertainty around a deterministic ArchesWeather forecast using flow matching. U-Cast combines a convolutional U-Net with Monte Carlo dropout and a deterministic-to-probabilistic training curriculum. Together, these systems represent distinct approaches to generating probabilistic weather forecasts.

2.1 Evaluation and metrics

IFS-ENS is evaluated with 50 ensemble members, whereas FGN, U-Cast and ArchesWeatherGen are evaluated with 64 members. For IFS-ENS, FGN and U-Cast, forecasts are available at lead times of 12, 48, 120, 168 and 240 h. ArchesWeatherGen is evaluated at the available lead times, but lacks 12-hourly output. IFS-ENS is verified against HRES-fc0, the deterministic ECMWF forecast initialized concurrently with the ensemble, whereas the AI systems are verified against ERA5. The following sections describe the metrics used to assess point-forecast, record-occurrence and distributional skill.

2.1.1 Point forecast

An ensemble forecast with K members for variable x at grid cell s, initialization time t_0, and lead time τ is

$$\left\{\hat{x}^{\tau,1}_{s,t_0}, \ldots, \hat{x}^{\tau,K}_{s,t_0}\right\}. \tag{5}$$

It can be collapsed to a point forecast using the ensemble mean,

$$\bar{x}^{\tau}_{s,t_0} = \frac{1}{K}\sum_{k=1}^{K} \hat{x}^{\tau,k}_{s,t_0}, \tag{6}$$

or the ensemble median. This reduction is useful for comparison with deterministic benchmarks, but it discards the predictive distribution and can be misleading because ensemble means are often smoother than individual members.

For an evaluation set I, the latitude-weighted RMSE of a point forecast $\tilde{x}^{\tau}_{s,t_0}$ is

$$\mathrm{RMSE}_I(\tau) = \left\{ \frac{\sum_{(s,t_0)\in I} \omega_s \left(\tilde{x}^{\tau}_{s,t_0} - x_{s,t_0+\tau}\right)^2}{\sum_{(s,t_0)\in I} \omega_s} \right\}^{1/2}, \tag{7}$$

where ω_s is the latitude weight. The corresponding bias is

$$\mathrm{Bias}_I(\tau) = \frac{\sum_{(s,t_0)\in I} \omega_s \left(\tilde{x}^{\tau}_{s,t_0} - x_{s,t_0+\tau}\right)}{\sum_{(s,t_0)\in I} \omega_s}. \tag{8}$$

We evaluate these scores on all valid grid-time pairs and on the observed-record sets.

2.1.2 Occurence probabilities

For upper-tail records, the ensemble probability of breaking the historical record is

$$\hat{p}^{\tau,+}_{s,t_0} = \frac{1}{K}\sum_{k=1}^{K} \mathbb{1}\left\{\hat{x}^{\tau,k}_{s,t_0} > r^{x,+}_{s,m(t_0+\tau)}\right\}. \tag{9}$$

For lower-tail records, the inequality is reversed. The occurrence problem is then a rare-event probabilistic classification problem: the forecast is $\hat{p}^{\tau}_{s,t_0}$ and the observation is $y^{\tau}_{s,t_0} \in \{0, 1\}$.

For an evaluation set I, the weighted observed and expected record counts are

$$N_I^{\tau} = \sum_{(s,t_0)\in I} \omega_s y^{\tau}_{s,t_0}, \qquad \hat{N}_I^{\tau} = \sum_{(s,t_0)\in I} \omega_s \hat{p}^{\tau}_{s,t_0}. \tag{10}$$

The difference $\hat{N}_I^{\tau} - N_I^{\tau}$ diagnoses whether records are over- or underforecast in expectation.

2.1.3 Precision, recall, and the forecaster's dilemma

For a probability cutoff c, a record alarm is

$$\hat{y}^{\tau}_{s,t_0}(c) = \mathbb{1}\left\{\hat{p}^{\tau}_{s,t_0} > c\right\}. \tag{11}$$

Precision and recall are

$$\text{Precision}(c) = \frac{\text{TP}(c)}{\text{TP}(c) + \text{FP}(c)}, \qquad \text{Recall}(c) = \frac{\text{TP}(c)}{\text{TP}(c) + \text{FN}(c)}. \tag{12}$$

Precision-recall curves are preferable to ROC curves for these rare events because true negatives dominate the global grid-time sample. In an early-warning setting, the high-precision, moderate-recall part of the curve is often especially relevant: a forecast that catches many records only by issuing alarms nearly everywhere is practically useless even if recall is high.

Because record-conditioned scoring selects on the observed outcome, it must be interpreted alongside full-sample occurrence diagnostics and forecast-conditioned checks. This is the practical implication of the forecaster's dilemma [Holzmann and Eulert, 2014, Lerch et al., 2017]: evaluating only observed extremes can reward overforecasting. In this work, record-conditioned scores are therefore treated as stress tests rather than standalone decision rules.

2.1.4 Brier score and reliability

The Brier score evaluates the numerical record probability directly [Brier, 1950]:

$$\text{BS}_I(\tau) = \frac{\sum_{(s,t_0)\in I} \omega_s \left(\hat{p}^{\tau}_{s,t_0} - y^{\tau}_{s,t_0}\right)^2}{\sum_{(s,t_0)\in I} \omega_s}. \tag{13}$$

The Murphy decomposition separates the Brier score into uncertainty, reliability, and resolution components [Murphy, 1973].

Reliability diagrams group predictions into probability bins and compare the mean predicted probability $\bar{p}_b$ with the observed event frequency $\bar{y}_b$. The current bins are deliberately small near zero:

$$[0, 0.001), [0.001, 0.01), [0.01, 0.02), [0.02, 0.05), [0.05, 0.10), \tag{14}$$

$$[0.10, 0.20), [0.20, 0.40), [0.40, 0.60), [0.60, 0.80), [0.80, 1]. \tag{15}$$

This resolution near zero is important because a probabilistic soft cap would appear most clearly when observed records receive zero or near-zero forecast probability.

2.1.5 Zero-probability miss rate

Reliability at $\hat{p} = 0$ can look deceptively good because record events are rare. The more operationally relevant quantity for the soft-cap question is the reverse conditional probability

$$\Pr(\hat{p}^{\tau} = 0 \mid y^{\tau} = 1), \tag{16}$$

the fraction of observed records for which no ensemble member crossed the historical record. Equivalently,

$$\Pr(\hat{p}^{\tau} = 0 \mid y^{\tau} = 1) = 1 - \mathrm{Recall}(c = 0). \tag{17}$$

This metric directly asks whether the event was inside the ensemble support at all. It is especially useful for distinguishing an ensemble that is miscalibrated but at least assigns nonzero tail probability from one that completely misses the observed record.

2.1.6 Continuous ranked probability score

The continuous ranked probability score is a proper scoring rule for real-valued probabilistic forecasts [Gneiting and Raftery, 2007, Jordan et al., 2019]. For predictive distribution F^{τ}_{s,t_0} and observation $x_{s,t_0+\tau}$, it is given by

$$\mathrm{CRPS}\left(F^{\tau}_{s,t_0}, x_{s,t_0+\tau}\right) = \int_{-\infty}^{\infty} \left(F^{\tau}_{s,t_0}(z) - \mathbb{1}\{x_{s,t_0+\tau} \le z\}\right)^2 \, dz. \tag{18}$$

It can be interpreted as an integral of threshold-specific Brier scores [Hersbach, 2000]. For an empirical ensemble with K members,

$$\widehat{\mathrm{CRPS}}^{\tau}_{s,t_0} = \frac{1}{K}\sum_{k=1}^{K}\left|\hat{x}^{\tau,k}_{s,t_0} - x_{s,t_0+\tau}\right| - \frac{1}{2K^2}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\left|\hat{x}^{\tau,k}_{s,t_0} - \hat{x}^{\tau,\ell}_{s,t_0}\right|. \tag{19}$$

The first term rewards ensemble members close to the verification, and the second term accounts for ensemble spread. In the deterministic limit, CRPS reduces to the mean absolute error.

2.1.7 Record margin and tail-coverage

For upper-tail records, define the observed record margin as

$$e^{+}_{s,t} = x_{s,t} - r^{x,+}_{s,m(t)}. \tag{20}$$

For lower-tail records,

$$e^{-}_{s,t} = r^{x,-}_{s,m(t)} - x_{s,t}. \tag{21}$$

For a margin threshold u, diagnostics are computed over records with $e_{s,t} > u$. If an ensemble is subject to a probabilistic soft cap, bias and out-of-ensemble rates should worsen as u increases.

In addition to CRPS, RMSE, and bias as functions of u, the margin outputs store the mean predicted record probability, the out-of-ensemble rate, and 90% ensemble coverage. For an upper record, the out-of-ensemble indicator is

$$\mathbb{1}\left\{x_{s,t_0+\tau} > \max_{1\le k\le K} \hat{x}^{\tau,k}_{s,t_0}\right\}, \tag{22}$$

with the inequality reversed for lower records. The 90% coverage diagnostic checks whether the verifying value falls between the empirical 5th and 95th percentiles of the ensemble. Together these diagnostics show whether the observed record is not merely above the threshold but represented at the observed intensity within the predictive distribution.

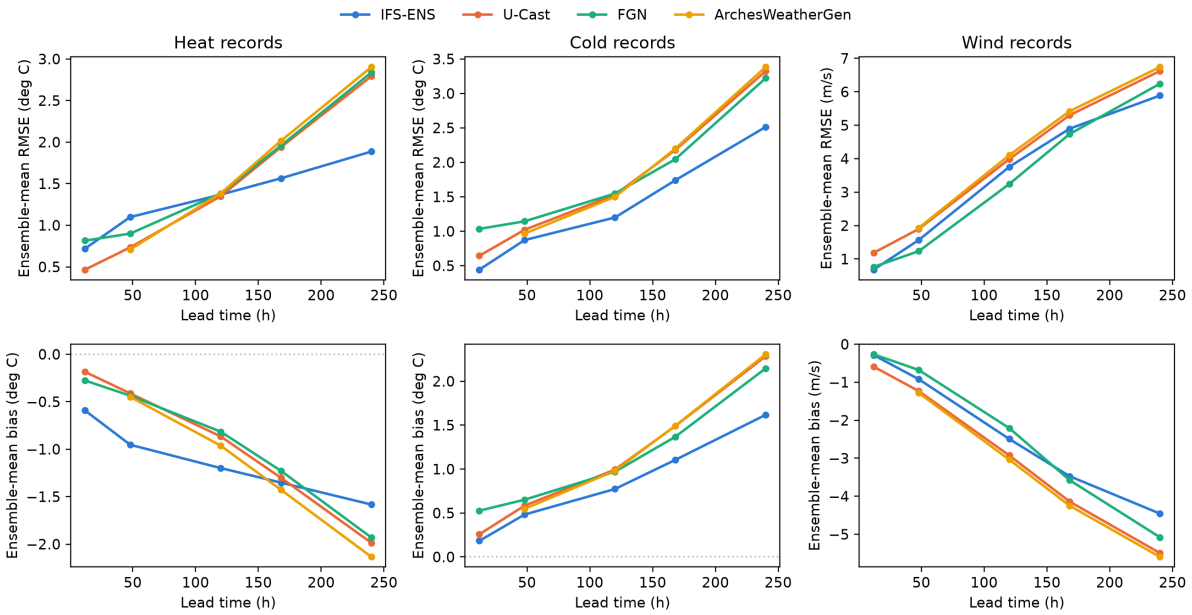


Figure 1: All-domain, all-season ensemble-mean point skill for 2022 record events. Rows show RMSE and bias; columns show heat, cold, and wind records. Negative heat and wind bias and positive cold bias correspond to attenuation of record intensity.

3 Results

3.1 Ensemble-mean point skill shows attenuation of record intensity

We first reduce each ensemble to its mean to enable comparison with the deterministic verification framework of Zhang et al. [2026]. Across all-domain, all-season record events, RMSE increases with lead time, while biases consistently attenuate event intensity: heat and wind records are forecast too cold and too weak, respectively, whereas cold records are forecast too warm (Figure 1). This pattern occurs in both physics-based and AI ensembles, with magnitudes that vary by event type and lead time. Conditioning on season and land–sea reveals substantial differences in terms of performance. In particular, long-lead biases are amplified for JJA land heat records and DJF land cold records (Supplementary Figures 8–9).

3.2 AI ensembles improve record-occurence forecasting

We now move to probabilistic evaluation and evaluate the full ensemble rather than its mean. The expected-minus-observed counts (Figure 2) and Brier scores (Figure 3) show that AI ensembles generally provide more accurate record probabilities than IFS-ENS, particularly for heat records. However, as mentioned before, note that the mixed verification targets complicate direct model ranking: IFS-ENS is verified against HRES-fc0, whereas the AI ensembles are verified against ERA5.

Considering a 48 h lead, precision–recall curves in the top panel of Figure 4 show that the AI ensembles generally have higher precision and recall than IFS-ENS, with the largest advantage for heat records. The bottom panel of Figure 4 shows reliability diagrams for the same 48 h lead. The AI ensembles are generally better calibrated than IFS-ENS, however, with a notable overconfidence, in particular for wind records. Similar results are obtained for 120 h, 168 h and

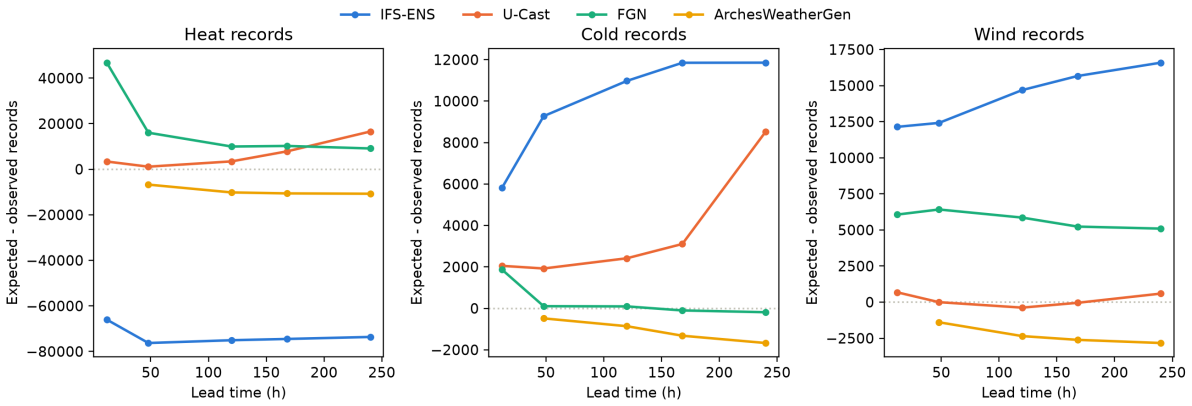


Figure 2: All-domain, all-season expected-minus-observed record counts. Positive values indicate that the ensemble assigns too much record probability; negative values indicate too little.

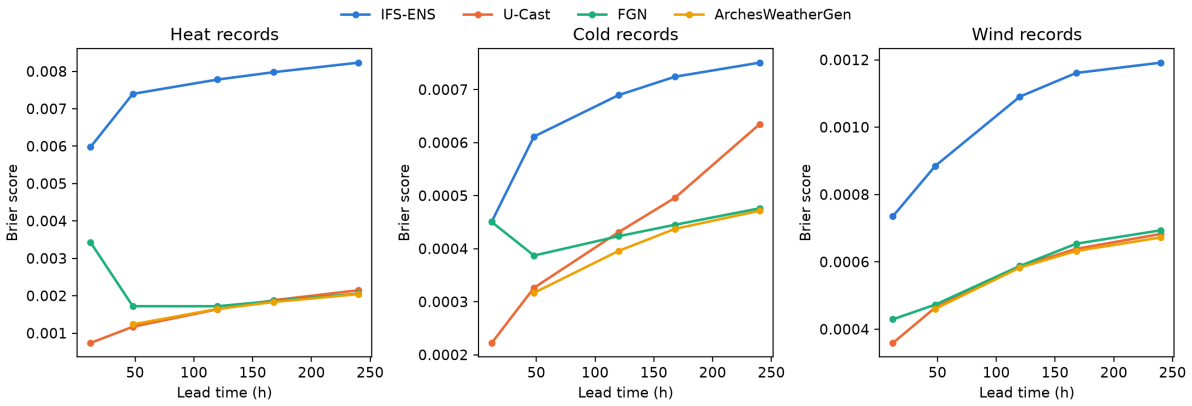


Figure 3: All-domain, all-season Brier scores for record occurrence. Lower values indicate more accurate probabilistic forecasts.

240 h leads (Supplementary Figures 10–12).

When it come to CRPS, record-conditioned CRPS (solid line) is substantially larger than all-weather CRPS (dashed line) for every event type and system, especially for wind at long lead (Figure 5). Record extremes are therefore a distinct and harder probabilistic verification regime.

Finally, the zero-probability miss rate reveals whether observed records fall outside the ensemble support altogether (Table 1). For heat records, the AI ensembles generally assign nonzero probability more often than IFS-ENS: at 48 h, IFS-ENS assigns zero probability to 47.5% of observed heat records, compared with 5.4%, 9.1% and 9.0% for U-Cast, FGN and ArchesWeatherGen, respectively. This pattern is not uniform across event types. IFS-ENS has lower zero-probability miss rates for cold records, whereas FGN remains particularly effective at retaining support for wind records at longer lead times. Zero-probability misses increase for several models with lead time, especially for wind, showing that nonzero threshold support remains difficult even where aggregate occurrence scores are favourable.

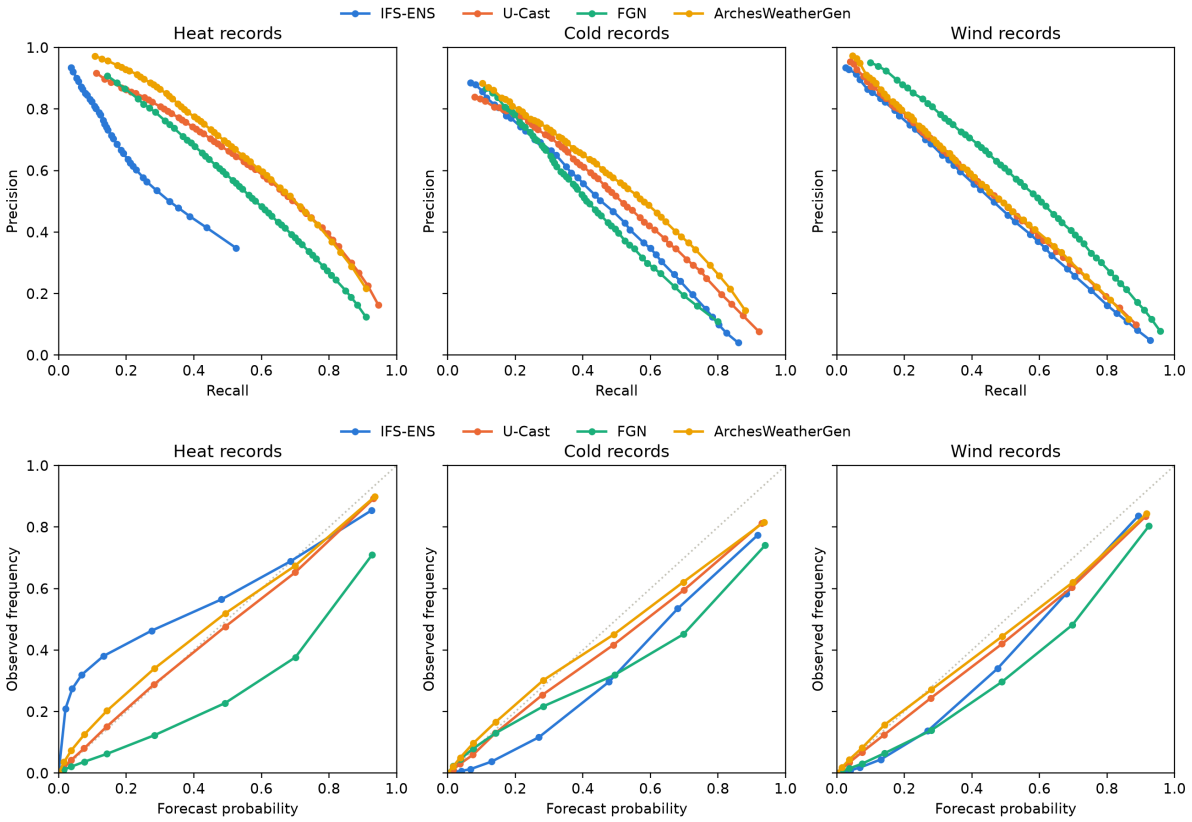


Figure 4: All-domain, all-season occurrence discrimination and calibration at 48 h. Top: precision–recall curves for probability-threshold alarms. Bottom: reliability diagrams; the diagonal denotes perfect calibration.

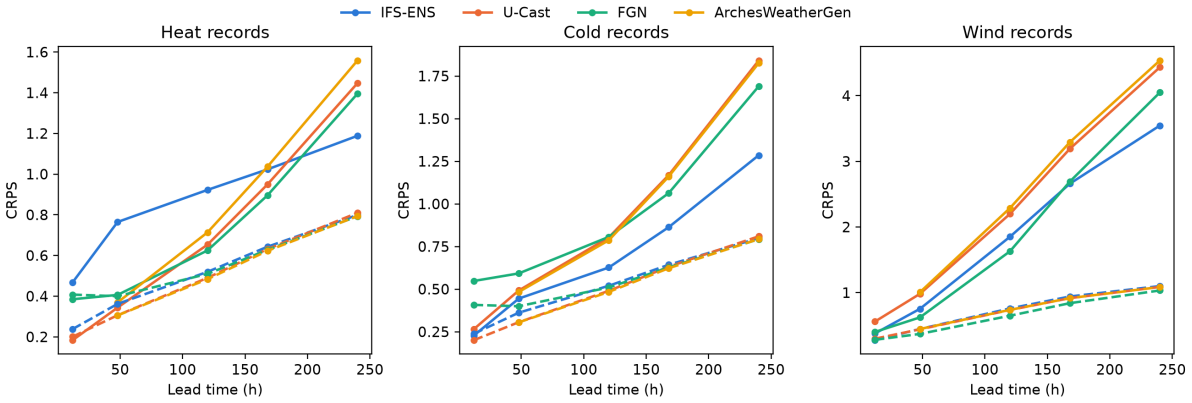


Figure 5: All-domain, all-season CRPS. Solid curves are conditioned on observed heat, cold, or wind records; dashed curves are all-weather CRPS. Lower values are better.

3.3 Forecasting record occurrence does not ensure accurate record intensity

Margin-conditioned diagnostics provide the most direct assessment of whether ensemble forecasts represent the intensity of the most unprecedented events. As the observed record margin increases,

Table 1: Zero-probability miss rate, $\Pr(\hat{p}^\tau = 0 \mid y^\tau = 1)$, for all-domain, all-season observed record events. The rate is the unweighted fraction of observed record grid-time pairs for which no ensemble member crossed the local historical record; lower values are better.

Record type	Model	12 h	48 h	120 h
Heat	IFS-ENS	0.282	0.475	0.448
Heat	U-Cast	**0.016**	**0.054**	0.138
Heat	FGN	0.120	0.091	**0.105**
Heat	ArchesWeatherGen	–	0.090	0.176
Cold	IFS-ENS	0.055	0.140	**0.172**
Cold	U-Cast	**0.025**	**0.079**	0.187
Cold	FGN	0.249	0.201	0.234
Cold	ArchesWeatherGen	–	0.120	0.205
Wind	IFS-ENS	0.042	0.072	0.171
Wind	U-Cast	0.060	0.113	0.247
Wind	FGN	**0.038**	**0.042**	**0.119**
Wind	ArchesWeatherGen	–	0.136	0.260

record-conditioned CRPS and ensemble-mean errors generally increase, with biases remaining directed towards less extreme values (Figure 6). At the same time, ensembles can retain substantial mean probability of record occurrence while empirical 90% coverage declines and out-of-ensemble rates increase (Figure 7).

For heat records exceeding the historical record by 4 K at 120 h, the AI ensembles assign mean record probabilities of approximately 0.77–0.85, yet their empirical 90% coverage is only 0.13–0.30. Thus, an ensemble may correctly anticipate that the historical threshold will be exceeded while failing to extend sufficiently far into the realised tail. The largest-margin cold and wind subsets contain relatively few events; we therefore interpret the corresponding curves as stress tests of tail support rather than as precise model rankings.

Together, these results distinguish two requirements for credible probabilistic forecasting of record extremes. First, the forecast must assign appropriate probability to record occurrence. Second, it must provide sufficient distributional support for the realised event intensity. The AI ensembles generally improve record-occurrence forecasting, particularly for heat, but this advantage does not consistently extend to conditional intensity skill or tail coverage. IFS-ENS remains competitive for cold and wind record intensity, especially at longer lead times, within the present evaluation.

4 Discussion

Record-breaking heat, cold and wind provide a demanding test of probabilistic weather forecasts because they lie beyond the local, month-specific historical range. The results show that record events are harder to forecast than average weather: record-conditioned CRPS exceeds all-weather CRPS across all evaluated systems and event types, with the largest degradation for wind at longer lead times. Performance also varies substantially by event type, lead time, season and land–sea class, indicating that record extremes cannot be treated as a single homogeneous verification problem.

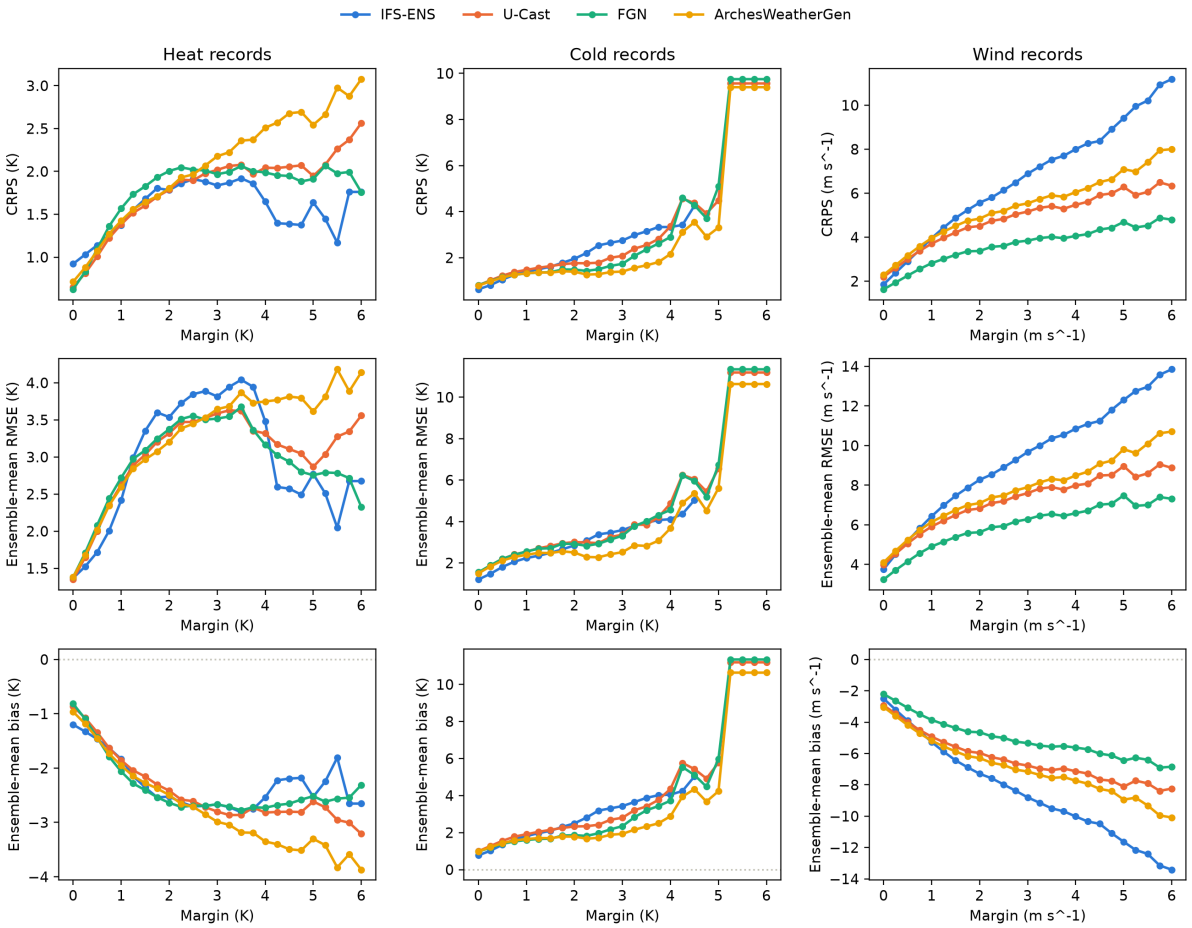


Figure 6: All-domain, all-season margin-conditioned intensity diagnostics at 120 h. Rows show CRPS, ensemble-mean RMSE, and ensemble-mean bias for records exceeding each margin threshold.

The full ensemble improves the assessment beyond ensemble-mean errors. AI ensembles generally produce more accurate probabilities of record occurrence than IFS-ENS, especially for heat. At 48 h, for example, U-Cast, FGN and ArchesWeatherGen assign zero probability to only 5.4%, 9.1% and 9.0% of observed heat records, respectively, compared with 47.5% for IFS-ENS. However, this advantage is not uniform across variables or metrics. IFS-ENS has lower zero-probability miss rates for cold records, while FGN retains the strongest threshold support for wind records at longer lead times. Reliability diagnostics further show that useful discrimination does not ensure calibrated probabilities, particularly for wind.

Most importantly, predicting record occurrence does not guarantee that the forecast distribution represents the realised record intensity. For 120-h heat records exceeding the previous record by 4 K, the AI ensembles retain mean record probabilities of approximately 0.77–0.85, but their empirical 90% coverage is only 0.13–0.30. Increasing record margins are generally accompanied by larger distributional errors, reduced coverage and more observations outside the ensemble range. These results show that probabilistic evaluation of record extremes should assess both occurrence probabilities and tail coverage at the realised intensity.

The analysis has several limitations. IFS-ENS is verified against HRES-fc0, whereas the AI

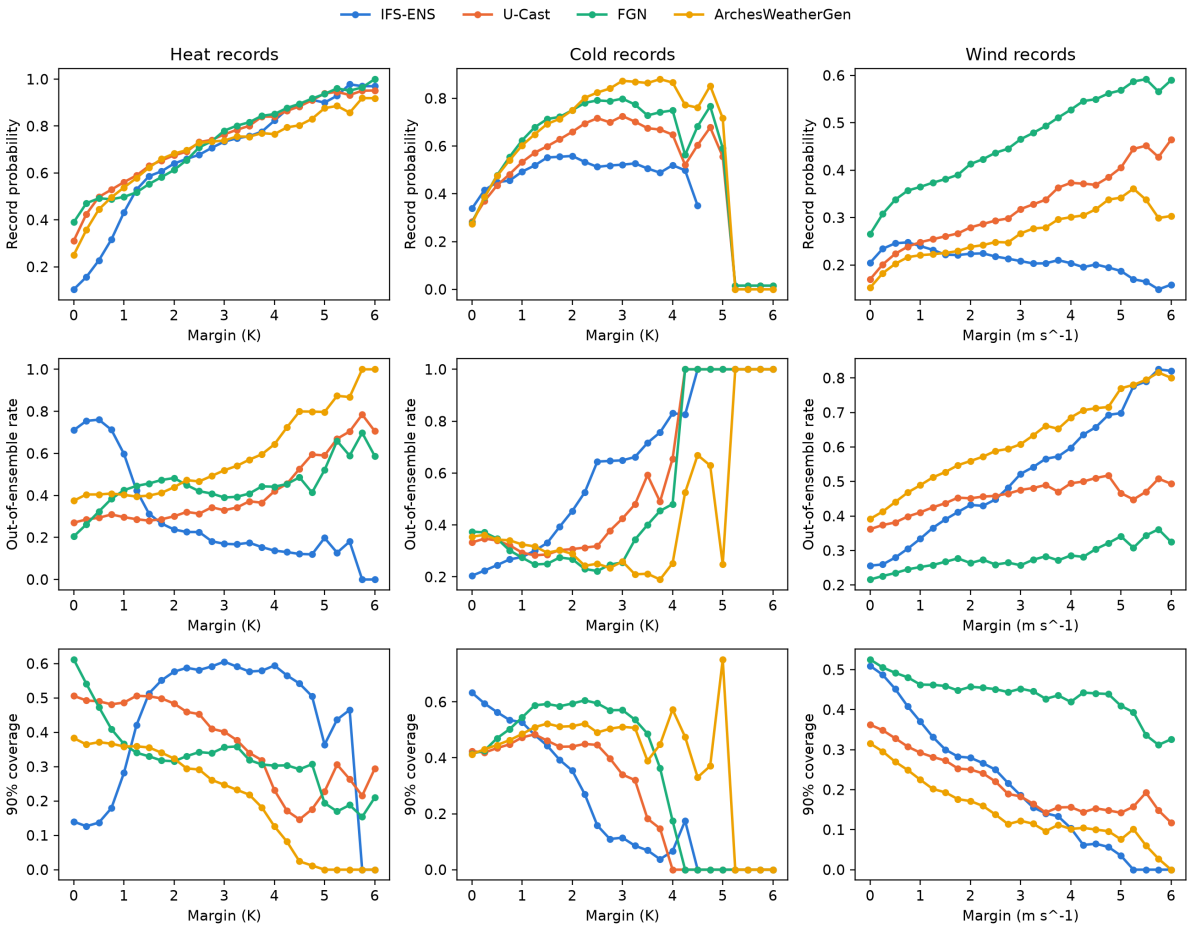


Figure 7: All-domain, all-season margin-conditioned probabilistic diagnostics at 120 h. Rows show mean record probability, out-of-ensemble rate, and empirical 90% coverage.

ensembles are verified against ERA5; direct cross-model comparisons therefore describe the present evaluation configurations rather than a common-target ranking. The analysis is restricted to 2022, and record events are spatially and temporally dependent, so the results do not establish multi-year robustness or uncertainty estimates. Priorities for future work are multi-year evaluation, sensitivity to a common verification target, and block- or event-based resampling to quantify uncertainty under spatiotemporal dependence.

5 Conclusions

We have extended the deterministic record-extreme benchmark from [Zhang et al., 2026] to probabilistic ensemble forecasts of record-breaking heat, cold and wind. Across 2022 forecasts from IFS-ENS, FGN, U-Cast and ArchesWeatherGen, record-conditioned CRPS is consistently larger than all-weather CRPS, confirming that record extremes are a distinct and more demanding verification regime. AI ensembles generally improve record-occurrence prediction, particularly for heat, but performance varies by event type and lead time. Calibration remains imperfect, especially for wind, and IFS-ENS remains competitive for cold and wind record intensity at longer lead times.

The results show that record occurrence and record intensity must be assessed separately. En-

sembles can assign substantial probability to exceeding a historical record while failing to cover the realised intensity of the event. Margin-conditioned diagnostics therefore complement occurrence scores by testing whether the predictive distribution extends sufficiently far into the observed tail. Record-breaking extremes should be adopted as a stress test for probabilistic AI weather forecasts, using diagnostics that jointly assess occurrence probabilities, conditional intensity and tail coverage.

Data and code availability

Acknowledgments

References

Ferran Alet, Tom R Andersson, Ilan Price, Stratis Markou, Andrew El-Kadi, Dominic Masters, Amy Li, Samier Merchant, Natalie Williams, Gregory Thornton, et al. Operational tropical cyclone forecasting with ai. *Nature*, pages 1–3, 2026.

Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian. Accurate medium-range global weather forecasting with 3d neural networks. *Nature*, 619:533–538, 2023. doi: 10.1038/s41586-023-06185-3.

Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078¡0001:VOFEIT¿2.0.CO;2.

Salva Rühling Cachay, Duncan Watson-Parris, and Rose Yu. U-cast: A surprisingly simple and efficient frontier probabilistic ai weather forecaster. *arXiv preprint arXiv:2604.09041*, 2026.

Lei Chen, Xiaohui Zhong, Feng Zhang, Yuan Cheng, Yinghui Xu, Yuan Qi, and Hao Li. FuXi: A cascade machine learning forecasting system for 15-day global weather forecast. *npj Climate and Atmospheric Science*, 6:190, 2023. doi: 10.1038/s41612-023-00512-1.

Copernicus Climate Change Service (C3S). Intense heatwave brings hottest june for western europe as the month ranks second warmest globally, July 2026. URL `https://climate.copernicus.eu/intense-heatwave-brings-hottest-june-western-europe-month-ranks` Accessed 2026-08-10.

Guillaume Couairon, Renu Singh, Anastase Charantonis, Christian Lessig, and Claire Monteleoni. Archesweathergen: Skillful and compute-efficient probabilistic weather forecasting with machine learning. *Science Advances*, 12(17):eadx2372, 2026.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Hans Hersbach. Decomposition of the continuous ranked probability score for ensemble prediction systems. *Weather and Forecasting*, 15(5):559–570, 2000. doi: 10.1175/1520-0434(2000)015¡0559:DOTCRP¿2.0.CO;2.

Hajo Holzmann and Matthias Eulert. The role of the information set for forecasting – with applications to risk management. *The Annals of Applied Statistics*, 8(1):595–621, 2014. doi: 10.1214/13-AOAS709.

Alexander Jordan, Fabian Krueger, and Sebastian Lerch. Evaluating probabilistic forecasts with scoringRules. *Journal of Statistical Software*, 90(12):1–37, 2019. doi: 10.18637/jss.v090.i12.

Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, Peter Wirnsberger, Meire Fortunato, Ferran Alet, Suman Ravuri, Timo Ewalds, Zach Eaton-Rosen, Weihua Hu, Alexander Merose, Stephan Hoyer, George Holland, Oriol Vinyals, Jacklynn Stott, Alexander Pritzel, Shakir Mohamed, and Peter Battaglia. Learning skillful medium-range global weather forecasting. *Science*, 382(6677):1416–1421, 2023. doi: 10.1126/science.adi2336.

Simon Lang, Mihai Alexe, Mariana CA Clare, Christopher Roberts, Rilwan Adewoyin, Zied Ben Bouallègue, Matthew Chantry, Jesper Dramsch, Peter D Dueben, Sara Hahner, et al. Aifs-crps: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score. *npj Artificial Intelligence*, 2(1):18, 2026.

Sebastian Lerch, Thordis L. Thorarinsdottir, Francesco Ravazzolo, and Tilmann Gneiting. Forecaster's dilemma: Extreme events and forecast evaluation. *Statistical Science*, 32(1):106–127, 2017. doi: 10.1214/16-STS588.

Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973. doi: 10.1175/1520-0450(1973)012¡0595:ANVPOT¿2.0.CO;2.

Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, et al. Probabilistic weather forecasting with machine learning. *Nature*, 637(8044):84–90, 2025.

Stephan Rasp, Stephan Hoyer, Alexander Merose, Ian Langmore, Peter Battaglia, Tyler Russell, Alvaro Sanchez-Gonzalez, Vivian Yang, Robert Carver, Shreya Agrawal, Matthew Chantry, Peter Dueben, Carla Bromberg, Jared Sisk, Luke Barrington, Aaron Bell, and Feng Sha. WeatherBench 2: A benchmark for the next generation of data-driven global weather models. *Journal of Advances in Modeling Earth Systems*, 16:e2023MS004019, 2024. doi: 10.1029/2023MS004019.

World Meteorological Organization. Western europe has hottest june on record, July 2026. URL `https://wmo.int/media/news/western-europe-has-hottest-june-record`. Accessed 2026-08-10.

Zhongwei Zhang, Erich Fischer, Jakob Zscheischler, and Sebastian Engelke. Physics-based models outperform ai weather forecasts of record-breaking extremes. *Science Advances*, 12(18):eaec1433, 2026.

Xiaohui Zhong, Lei Chen, Hao Li, Roberto Buizza, Jun Liu, Jie Feng, Zijian Zhu, Xu Fan, Kan Dai, Jing-jia Luo, et al. Fuxi-ens: A machine learning model for efficient and accurate ensemble weather prediction. *Science Advances*, 11(44):eadu2854, 2025.

A Supplementary materials

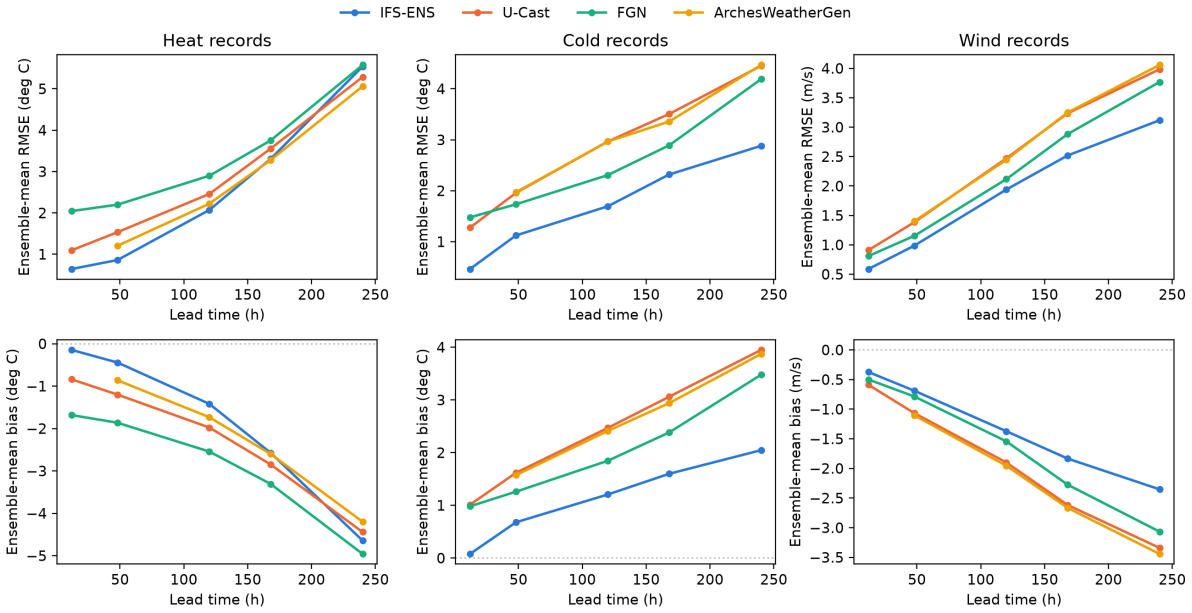


Figure 8: Ensemble-mean point skill for JJA land record events. Rows show RMSE and bias; columns show heat, cold and wind records. Negative heat and wind bias and positive cold bias indicate systematic underestimation of record magnitude.

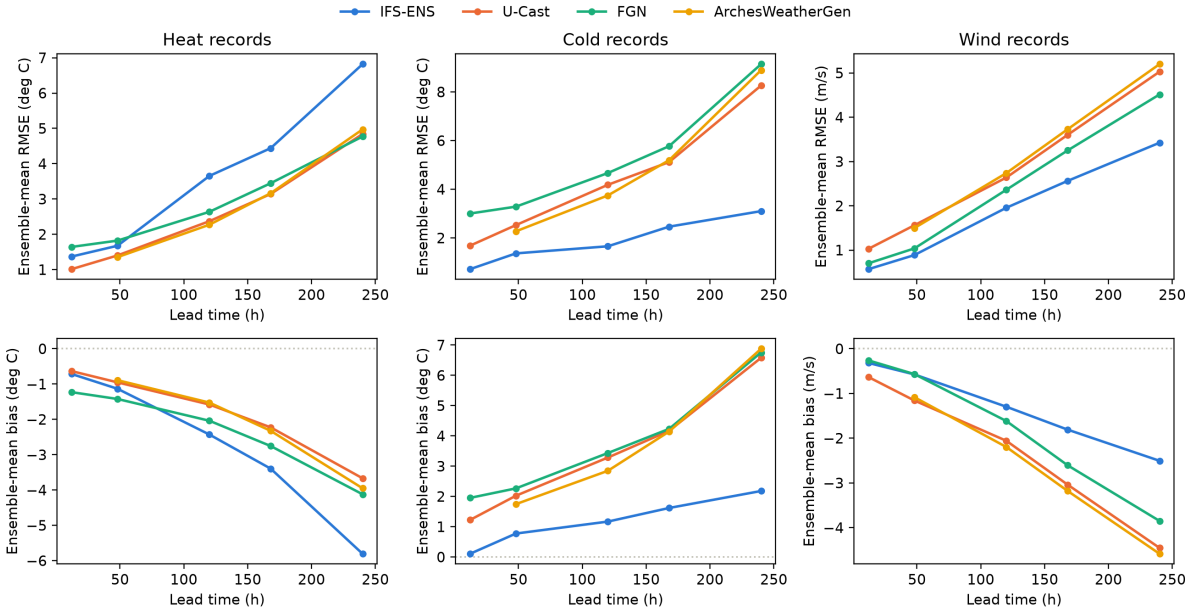


Figure 9: AsEnsemble-mean point skill for DJF land record events. Rows show RMSE and bias; columns show heat, cold and wind records. Negative heat and wind bias and positive cold bias indicate systematic underestimation of record magnitude.

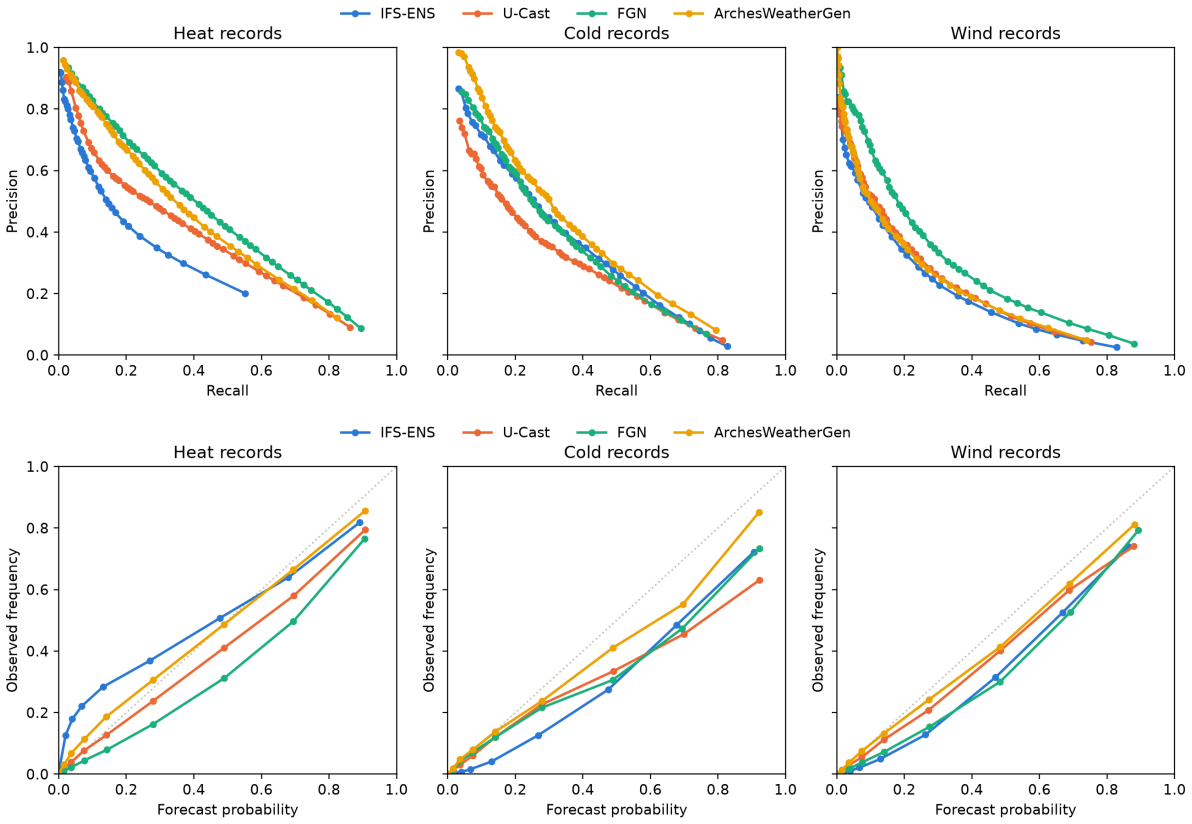


Figure 10: All-domain, all-season occurrence discrimination and calibration at 120 h. Top: precision–recall curves for probability-threshold alarms. Bottom: reliability diagrams; the diagonal denotes perfect calibration.

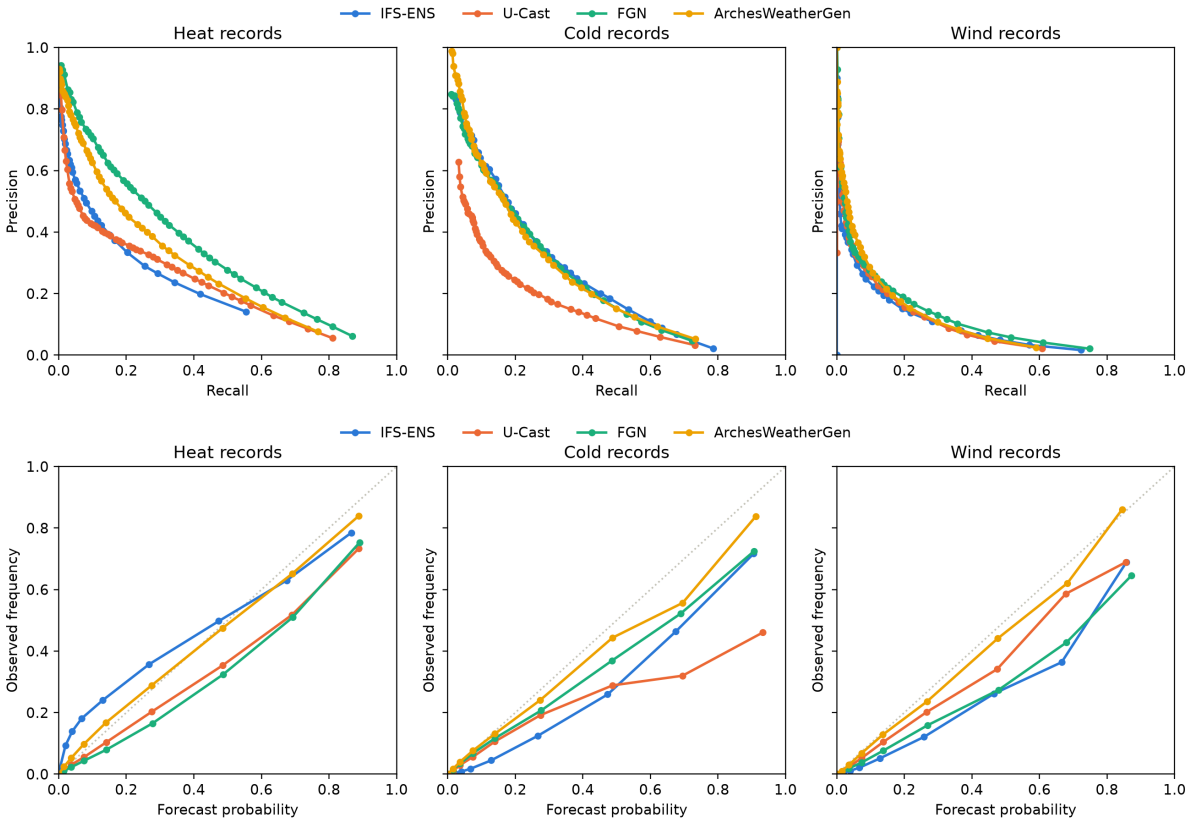


Figure 11: All-domain, all-season occurrence discrimination and calibration at 168 h. Top: precision–recall curves for probability-threshold alarms. Bottom: reliability diagrams; the diagonal denotes perfect calibration.

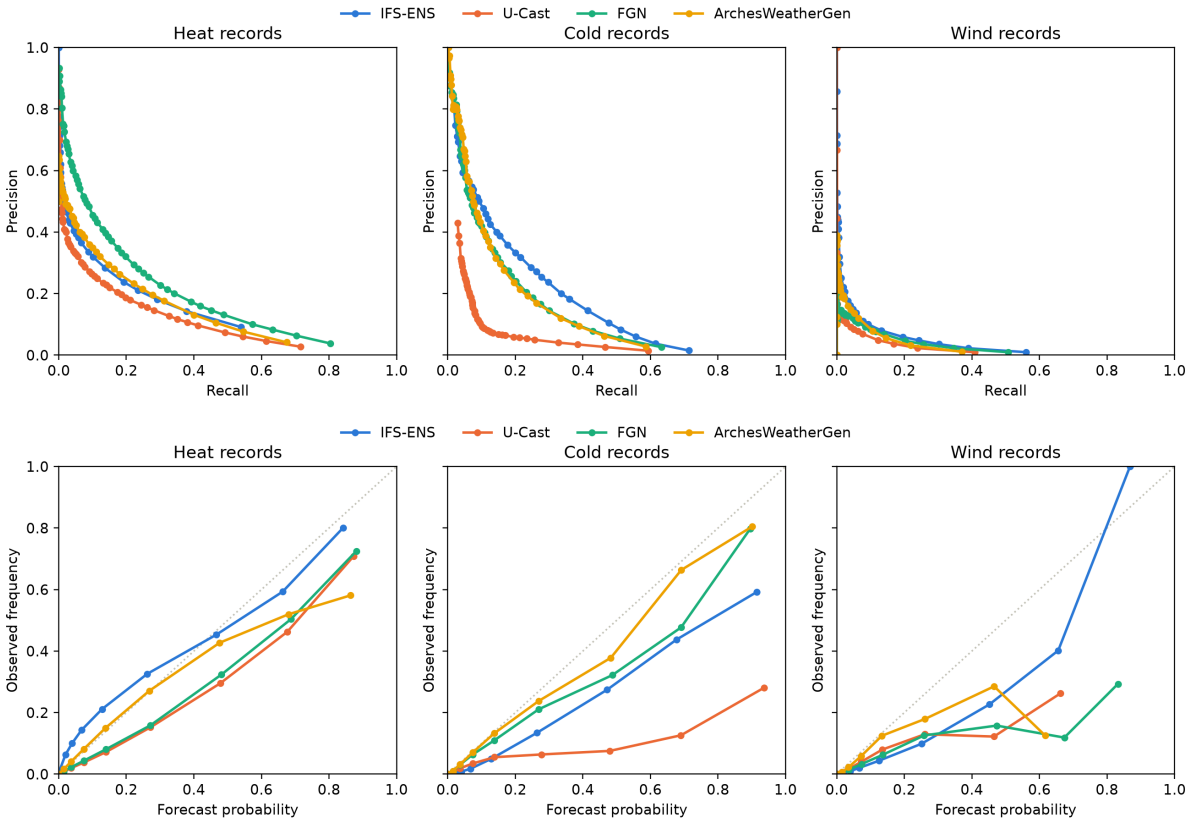


Figure 12: All-domain, all-season occurrence discrimination and calibration at 240 h. Top: precision–recall curves for probability-threshold alarms. Bottom: reliability diagrams; the diagonal denotes perfect calibration.

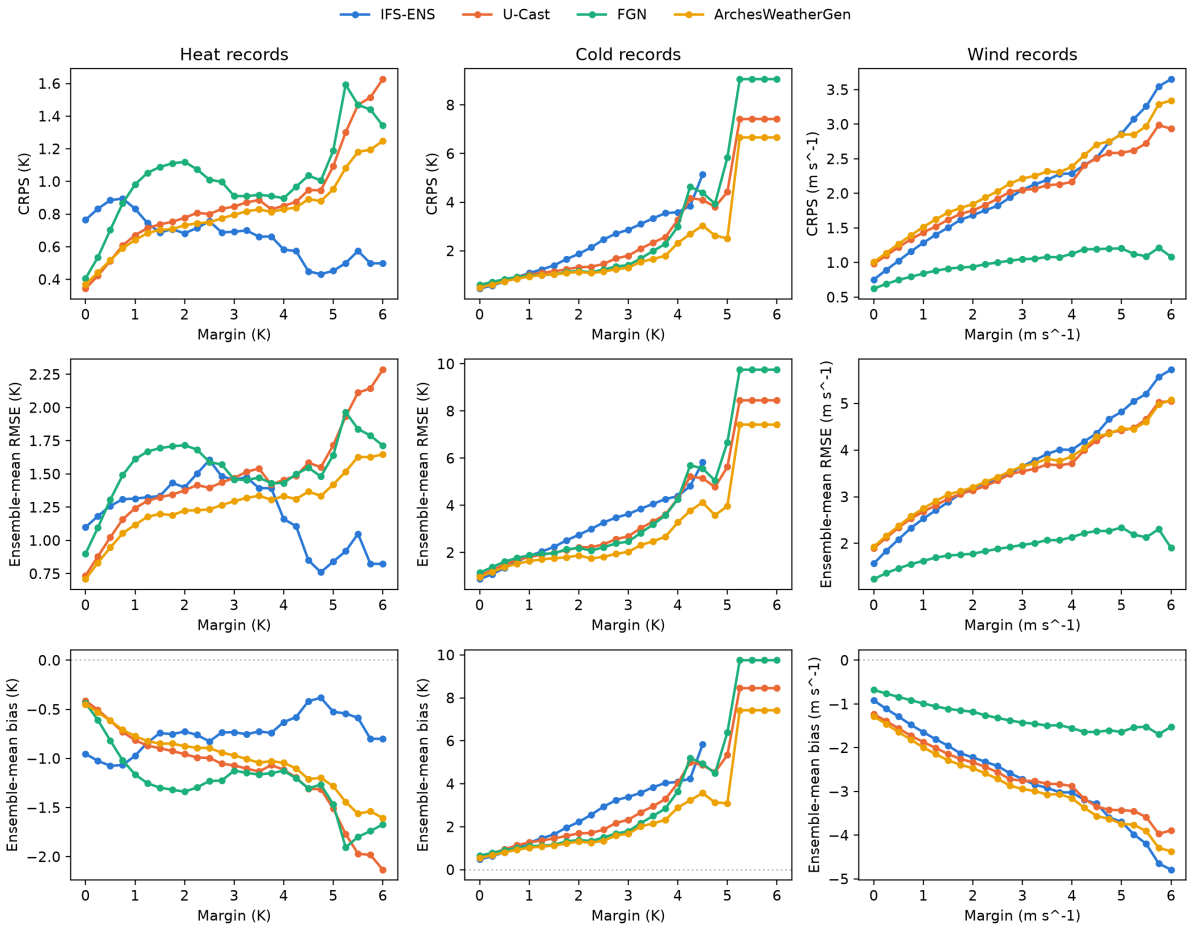


Figure 13: All-domain, all-season margin-conditioned intensity diagnostics at 48 h. Rows show CRPS, ensemble-mean RMSE, and ensemble-mean bias for records exceeding each margin threshold.

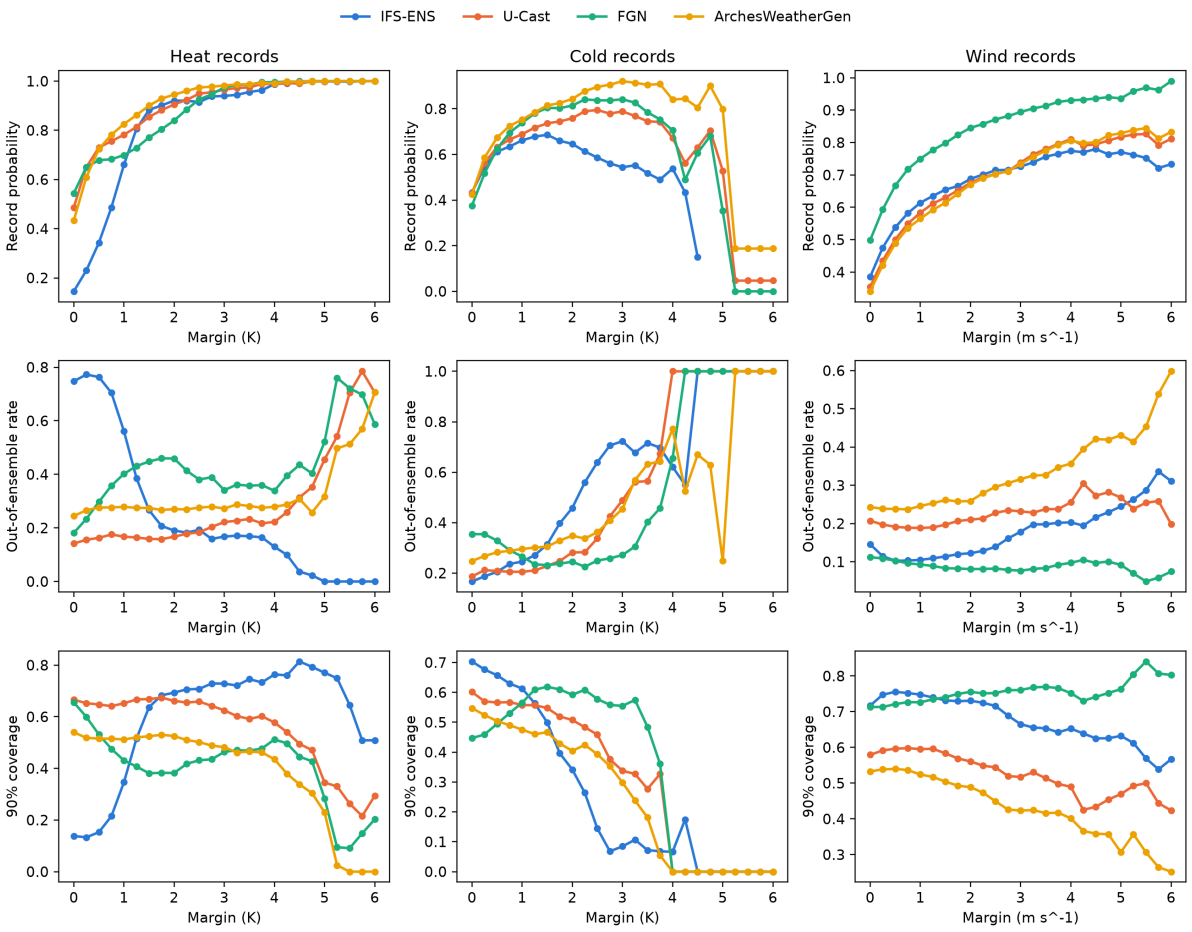


Figure 14: All-domain, all-season margin-conditioned probabilistic diagnostics at 48 h. Rows show mean record probability, out-of-ensemble rate, and empirical 90% coverage.

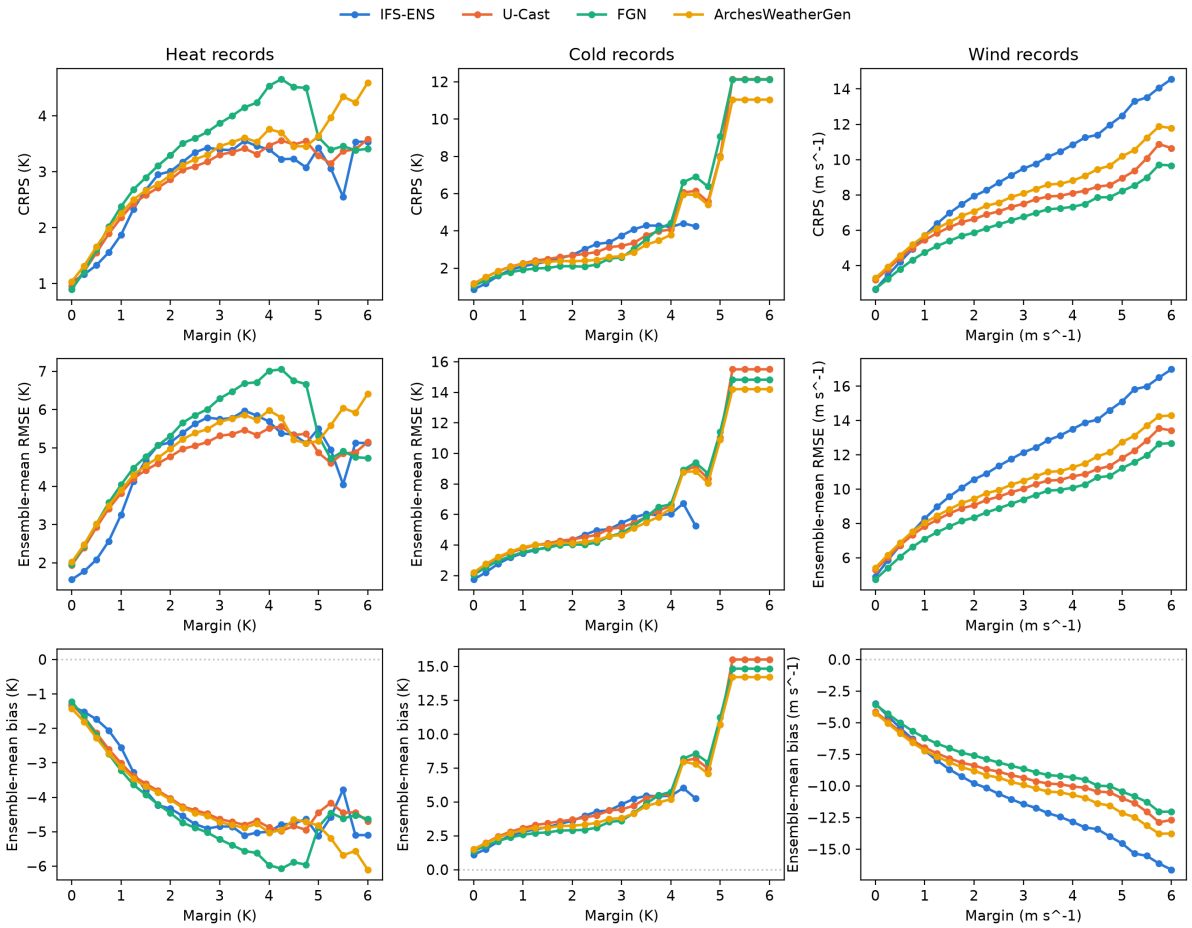


Figure 15: All-domain, all-season margin-conditioned intensity diagnostics at 168 h. Rows show CRPS, ensemble-mean RMSE, and ensemble-mean bias for records exceeding each margin threshold.

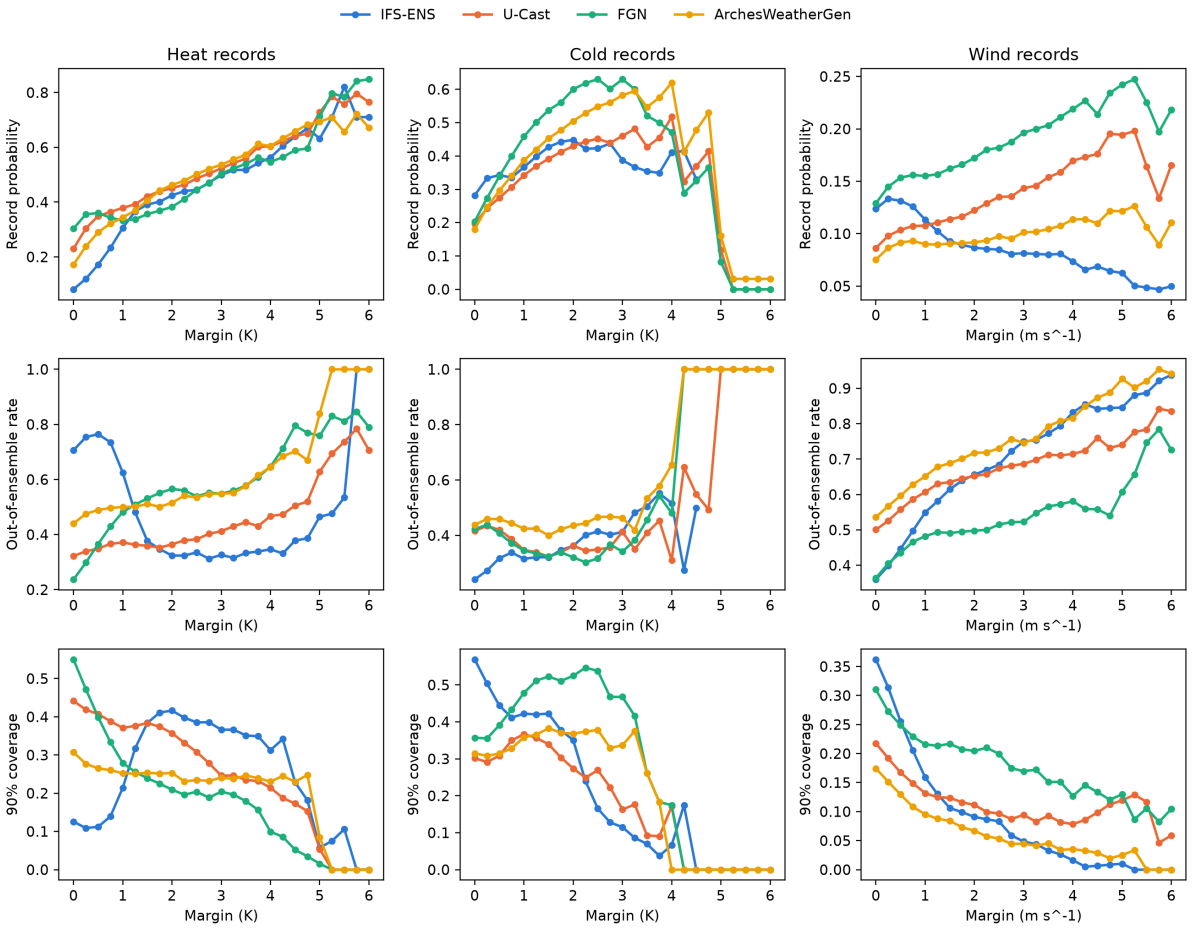


Figure 16: All-domain, all-season margin-conditioned probabilistic diagnostics at 168 h. Rows show mean record probability, out-of-ensemble rate, and empirical 90% coverage.

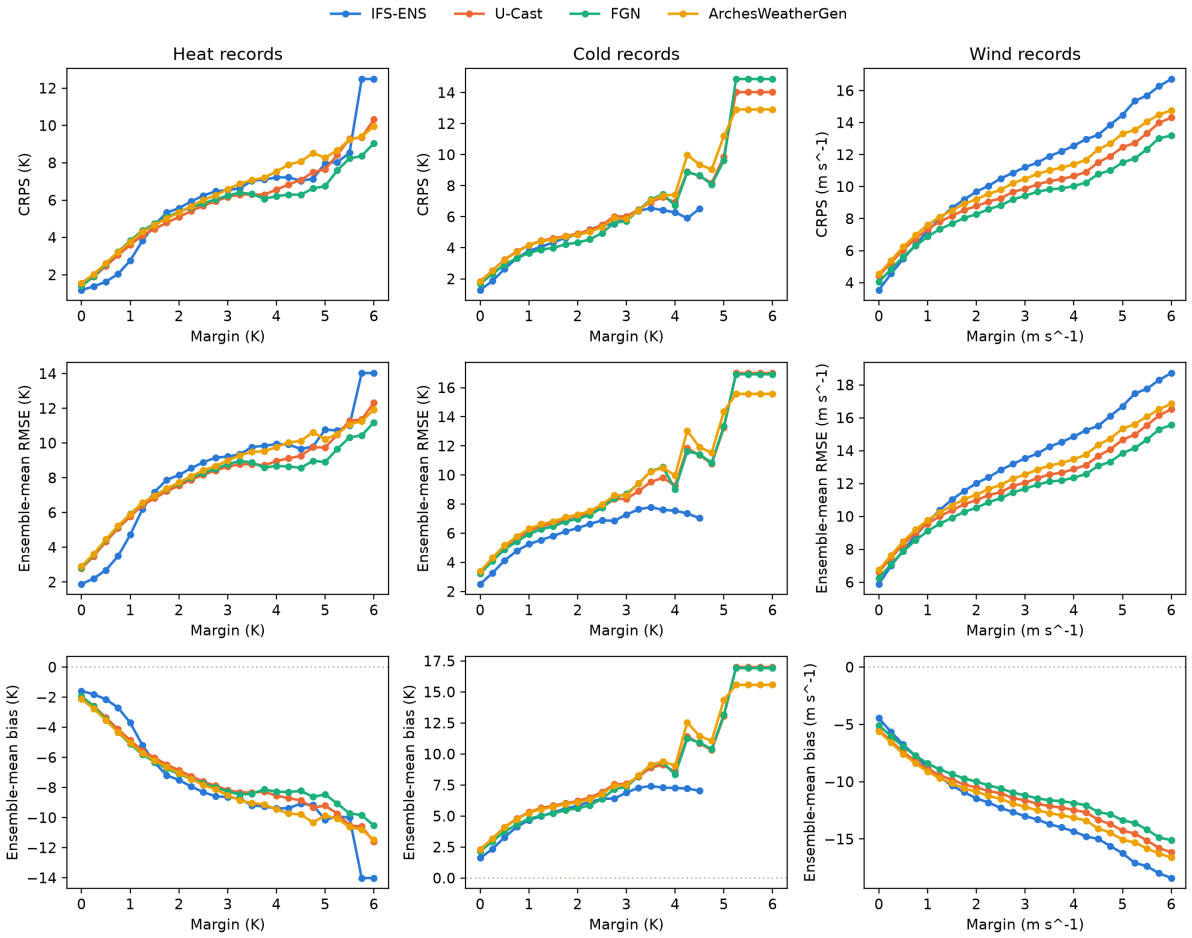


Figure 17: All-domain, all-season margin-conditioned intensity diagnostics at 240 h. Rows show CRPS, ensemble-mean RMSE, and ensemble-mean bias for records exceeding each margin threshold.

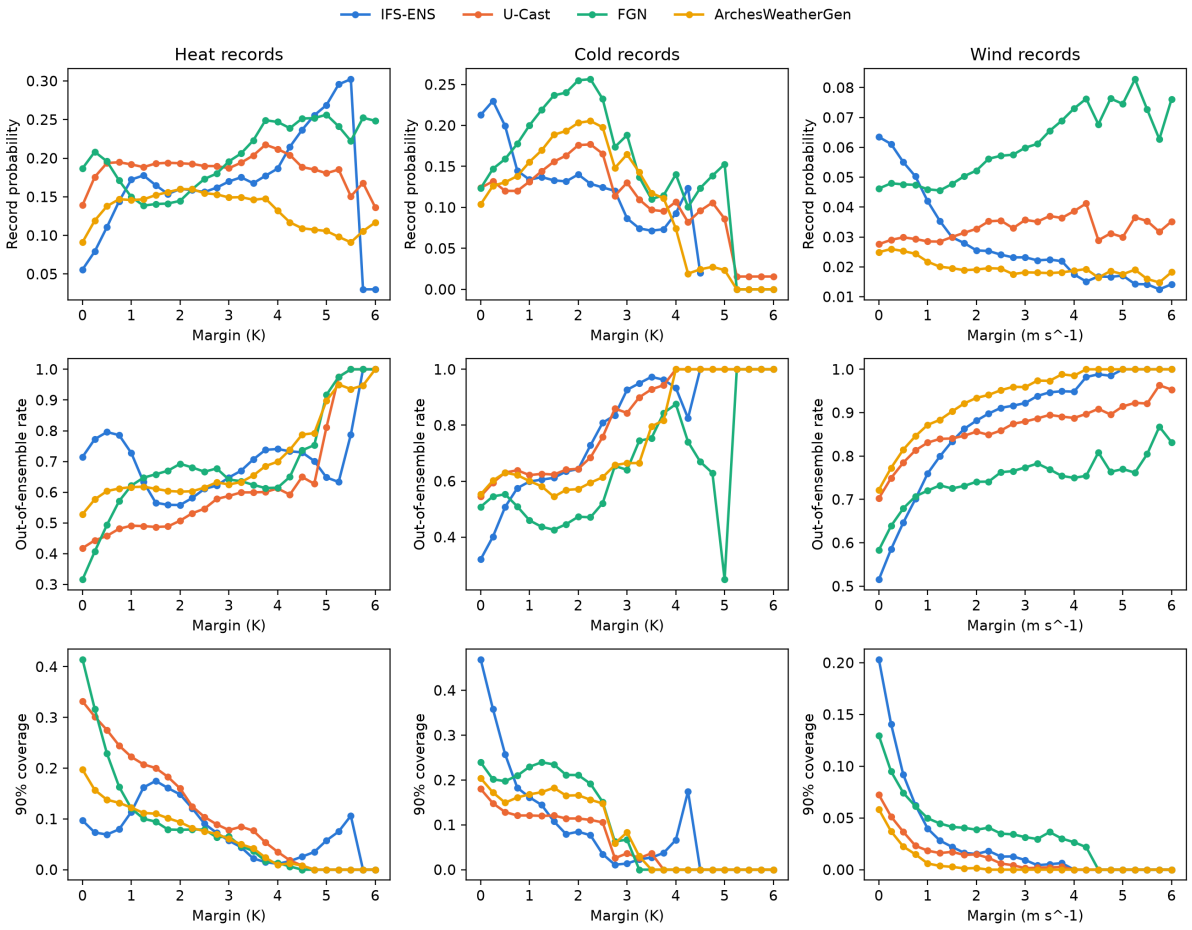


Figure 18: All-domain, all-season margin-conditioned probabilistic diagnostics at 240 h. Rows show mean record probability, out-of-ensemble rate, and empirical 90% coverage.